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Hybrid Radiomics and Machine Learning for Brain Tumors Multi-Task Classification: An Exploratory Study on Integrating GLCM and Curvelet-Based Features for Enhanced Accuracy

Mostafa Jafari[1] | Masoumeh Rahmani[2] | Sanaz Alibabaie[3] | Jafar Fatahiasl[1] | Sasan Razmjoo[4] | Mohammad Javad Tahmasebi Birgani[5] | Marziyeh Tahmasbi[1,6]

[1]Department of Radiologic Technology, School of Allied Medical Sciences, Ahvaz Jundishapur University of Medical Sciences, Ahvaz, Iran | [2]Department of Bioelectrics and Biomedical Engineering, School of Advanced Technologies in Medicine, Isfahan University of Medical Sciences, Isfahan, Iran | [3]Department of Medical Physics, Faculty of Medicine, Semnan University of Medical Sciences, Semnan, Iran | [4]Department of Clinical Oncology and Clinical Research Development Center, Golestan Hospital, Ahvaz Jundishapur University of Medical Sciences, Ahvaz, Iran | [5]Department of Medical Physics, Faculty of Medicine, Ahvaz Jundishapur University of Medical Sciences, Ahvaz, Iran | [6]Cancer Research Center, Ahvaz Jundishapur University of Medical Sciences, Ahvaz, Iran

Correspondence: Marziyeh Tahmasbi (marziyeh_tahmasbi@yahoo.com)

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ABSTRACT

Background and Aims: Accurate classification of brain tumors is vital for effective treatment planning. Manual assessment of magnetic resonance imaging (MRI) scans is often subjective and time-consuming. This exploratory study proposes a machine learning approach integrating radiomic features from contrast-enhanced T1-weighted MRI scans to improve classification accuracy for glioblastomas (GBM), meningiomas, brain metastases, and normal controls (NC).

Methods: A total of 154 patients (74 GBM, 62 meningioma, 18 brain metastases) and 170 NCs were included in the study. MRI scans underwent standard preprocessing. To address class imbalance, data augmentation was applied to the GBM, meningioma, and brain metastases groups, increasing their sample sizes to 148, 124, and 180 cases, respectively. Radiomic features were extracted using gray-level co-occurrence matrix (GLCM) analysis and Curvelet-based transformations. Feature selection was performed using the Least Absolute Shrinkage and Selection Operator (LASSO), followed by multicollinearity assessment using the Spearman correlation test. Principal component analysis (PCA) was also performed. Nine machine learning classifiers, including Random Forest (RF), K-Nearest Neighbors (KNN), Categorical Boosting (CatBoost), Extreme Gradient Boosting (XGBoost), Support Vector Machine (SVM), Logistic Regression (LR), Naive Bayes, AdaBoost, and LightGBM, were trained and evaluated.

Results: Random Forest and CatBoost achieved the highest performance, with 95.2% accuracy using LASSO-selected features, outperforming other classifiers. Combining GLCM and Curvelet features improved accuracy over either set alone. PCA-based dimensionality reduction (99% variance) also performed well, but slightly below LASSO.

Conclusion: Integrating Curvelet and GLCM-based radiomics with optimized models improves diagnostic accuracy, offering a promising tool to assist radiologists in tumor classification.

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1 | Introduction

Brain tumors are one of the most common life-threatening diseases of the central nervous system. A brain tumor, whether benign or malignant, represents an uncontrolled proliferation of cancerous cells within the brain. Benign tumors typically exhibit a more uniform structure devoid of actively dividing cells, whereas malignant tumors present a heterogeneous structure with active cellular proliferation [1]. Among the primary malignant brain tumors, meningiomas and gliomas are examples of benign brain tumors, while glioblastoma multiforme (GBM) stands out as one of the most aggressive and prevalent types [2].

Early tumor diagnosis is vital for effective treatment and recovery [3]. Various brain imaging techniques are used for this purpose, with magnetic resonance imaging (MRI) being the most common due to its ability to produce high-quality images of brain abnormalities [4]. Contrast-weighted T1 MRI sequences are particularly effective for detecting different brain abnormalities such as GBM tumors, brain metastases, and meningiomas, and monitoring treatment response, as they clearly outline tumor margins [5–7]. MRI is also essential for evaluating tumor progression or regression, with changes in contrast or signal intensity reflecting disease dynamics [5].

Evaluating MR images qualitatively and visually for tumor diagnosis and delineation is challenging due to variability stemming from reliance on clinician expertise and the associated extensive time demands [1, 8]. Radiomics, the high-throughput extraction of quantitative imaging features, addresses these challenges by objectively characterizing tumor heterogeneity beyond visual discernment [9]. Incorporating these radiomic features with machine learning models has demonstrated promise in enhancing the accuracy and precision of brain tumor detection, classification, and treatment evaluation, thus potentially improving clinical outcomes [2, 5, 9]. Quantitative methods frequently utilize image texture analysis, employing mathematical techniques to characterize the grayscale patterns and spatial distributions of intensity changes in MR images [5, 10, 11]. These texture features significantly aid automated image classification through machine learning algorithms, effectively facilitating the detection and grading of brain tumors [12].

Radiomics has significantly contributed to brain tumor analysis; however, current approaches still face notable limitations. While advanced deep-learning methods have shown promise in early detection and multi-class brain tumor classification [13], some existing studies focus on simpler binary classification problems, such as differentiating low-grade from high-grade gliomas or distinguishing tumor tissue from normal tissue [3, 14]. To effectively capture the intricate, multi-scale heterogeneity inherent in brain tumors, radiomic pipelines must incorporate deeper, higher-order features capable of enhancing predictive performance [15]. Such methodological limitations constrain the robustness and transferability of radiomic models, particularly under conditions involving varying MRI acquisition protocols or class imbalance, factors known to adversely affect feature reproducibility and consistency [16]. To overcome these limitations, this study introduces a hybrid radiomics framework integrating gray-level co-occurrence matrix (GLCM) based texture features with Curvelet-based multi-scale descriptors. Although prior studies have explored combinations of these features individually or partially [17, 18], their comprehensive integration specifically for multi-class brain tumor classification remains understudied. Our proposed framework concurrently captures spatial and frequency-domain tumor characteristics, providing a richer, more informative feature set that effectively addresses the subtle heterogeneity inherent to brain tumors, thereby enhancing classification accuracy and clinical applicability.

Given the crucial role of early and accurate diagnosis in improving outcomes for brain tumor patients, this study introduces a machine learning-based classification framework integrating GLCM-based and Curvelet-based radiomic features. The Least Absolute Shrinkage and Selection Operator (LASSO) algorithm [19] was utilized for initial feature selection, followed by Spearman rank correlation ($|\rho| > 0.70$) to identify and manage multicollinearity among features. Subsequently, principal component analysis (PCA) was applied for further dimensionality reduction [20]. This structured multi-step process optimizes the selection of the most informative features, enhancing classification accuracy. Various machine learning models were comprehensively evaluated to identify the most effective classifier for distinguishing glioblastomas, meningiomas, brain metastases, and normal brain tissues. Our integrated methodology not only demonstrates superior performance compared to existing radiomic methods but also offers a robust, interpretable platform readily integrable into clinical workflows, supporting radiologists in clinical decision-making. The main contributions of this study are summarized as follows:

- Integration of multiple radiomic features: Development of a hybrid radiomics model combining GLCM-based features and Curvelet-based multi-scale descriptors.
- Optimized feature selection: Application and comparison of LASSO and PCA to select the most informative features and reduce dimensionality.
- Enhanced diagnostic accuracy: Achieved superior accuracy (95.2%) in classifying multiple brain tumor types compared to existing single-feature radiomics models.
- Robust comparative analysis: Comprehensive evaluation of nine machine learning algorithms to determine the optimal model for multi-class brain tumor classification.
- Clinical relevance: Provision of an interpretable and clinically applicable tool aiding radiologists in accurate and efficient brain tumor diagnosis.

The remainder of this article is structured as follows. Section 2 presents a review of related work and highlights the research gap. Section 3 details the materials and methods, including data acquisition, preprocessing, radiomic feature extraction, feature selection, and classification algorithms. Section 4 reports the experimental results and model evaluations. Section 5 discusses the findings in comparison with existing literature and outlines clinical implications and study limitations. Finally, Section 6 concludes the study and suggests directions for future research.

2 | Related Works

Parekh et al. [21] developed a feature extraction algorithm with 93% sensitivity and perfect specificity (100%) for distinguishing between Grade II and Grade IV brain tumors. Similarly, Chen et al. [3] introduced a gradient-based classification method for glioma detection and grading, achieving 86.3% accuracy and 89.4% sensitivity for detection, and 76.30% accuracy with 83.70% sensitivity for grading. Additionally, Bahadure et al. [1] employed the Berkeley Wavelet Transformation (BWT) for segmentation and a support vector machine (SVM) for classification and analyzing MR images of 15 glioma patients. Their method achieved 96.51% accuracy, with 94.20% precision and 97.72% sensitivity. Furthermore, Cepeda et al. [22] highlighted the integration of MRI radiomic features with intraoperative ultrasound to predict meningioma consistency. They evaluated several classifiers, with Random Forest (RF) achieving the highest accuracy of 94%, followed by SVM (91%), K-Nearest Neighbors (KNN) (89%), and Logistic Regression (LR) (85%). This approach demonstrates the potential of radiomic analysis and machine learning to enhance pre-surgical planning and tumor characterization accuracy. These studies underscore the value of textural feature analysis in enhancing the precision of tissue characterization and tumor grading in brain imaging.

Numerous studies have applied machine learning and deep learning methods to brain tumor classification, with varying feature types and classification scopes. Malakouti et al. [23] achieved 95.7% accuracy using LightGBM trained on GLCM features for binary tumor detection, while Rehman Khan et al. [24] reported 95.1% using DenseNet169-based features with ensemble classifiers. Other convolutional neural network (CNN)-based models, such as those by Celik et al. [25] and Kibriya et al. [26] exceeded 97% accuracy but required large data sets and lacked interpretability. In contrast, Priya et al. [27] and Cepeda et al. [22] used handcrafted radiomic features and simpler classifiers for subtype differentiation but were limited in sample size or model complexity. These studies reveal gaps in generalizability, feature diversity, and classification granularity. Most notably, few have combined multiple texture descriptors, such as GLCM and Curvelet features, for multi-class classification using interpretable machine learning models. This underscores the need for hybrid approaches that integrate complementary features and robust classifiers to improve diagnostic performance while maintaining transparency.

3 | Materials and Methods

3.1 | Study Population and Data Collection

This retrospective study was approved by the institutional ethics committee (Ethics code: IR.AJUMS.REC.1399.839). Clinical records of brain tumor patients who had undergone treatment at Golestan Hospital (Ahvaz, Iran) from February 2021 to February 2024 were reviewed to determine eligible participants. The inclusion criteria included patients diagnosed with GBM, meningioma, or brain metastases, based on the guidelines of the World Health Organization (WHO) [28–30]. In addition, patients were included if their initial diagnosis MR images were available in the hospital's PACS system. Patients or their companions' informed consent was not needed because of the retrospective nature of the study. Based on these criteria, a total of 154 patients were included in the study, consisting of 74 patients with GBM, 62 with meningioma, and 18 with brain metastases. Furthermore, brain MR images from 170 NCs were also obtained from PACS. All study procedures adhered to the Declaration of Helsinki. Patient data confidentiality was strictly maintained, and all images were anonymized before use.

3.2 | MRI Acquisition

All MR images were acquired using a clinical 1.5 Tesla MRI system (Siemens, Germany). The imaging protocol included a series of conventional MRI (C-MRI) sequences. These sequences consisted of axial T1, T2, FLAIR, sagittal T2, and coronal T2, along with contrast-enhanced T1 sequences (axial, sagittal, and coronal). From these sequences, axial contrast-enhanced T1-weighted ones were selected as high-resolution anatomical images for feature extraction. The MR image acquisition parameters included TR/TE = 0.019, flip angle = 90°, and acquisition voxel size = $1.1 \times 0.9 \times 5.5\ mm^3$.

3.3 | Preprocessing and Augmentation of MR Images

For each evaluated patient, the image slice with the maximum cross-sectional area of the tumor from the images was selected by a radiation oncologist with enough work experience. For NCs also, the slice with the maximum brain area was selected. These DICOM images were imported into the scientific software package MATLAB (R2020, The Math Works Inc., Natick, MA, USA) to apply skull stripping using an in-house prepared code. Skull stripping was performed using a threshold-based binary segmentation (gray level > 20), followed by morphological opening to remove small artifacts, hole-filling to ensure tissue continuity, and boundary erosion with a disk-shaped structuring element (radius = 15) to refine the mask. The rest of the preprocessing steps were applied using Python (3.10.0). The resulting brain-extracted images were then resized to 256×256 pixels and discretized to 256 gray levels. Denoising was carried out using a bilateral filter, with the degree of smoothing set at twice the patch variance, effectively reducing noise while preserving key edge details. Upon completing preprocessing, data augmentation was employed to address limited sample sizes within the tumor groups and to improve class balance. Each brain metastasis image was augmented tenfold (increasing from 18 to 180), while glioblastoma (GBM) and meningioma images were each doubled (from 74 to 148 and from 62 to 124, respectively). The augmentation process involved random horizontal and vertical flips, along with in-plane rotations of ±10°. Normal control images were excluded from augmentation. Augmentation procedures were also implemented using Python scripts, resulting in a more balanced and comprehensive data set for subsequent feature extraction and classification. A visualization of the number of cases before and after augmentation is presented in Figure 1.

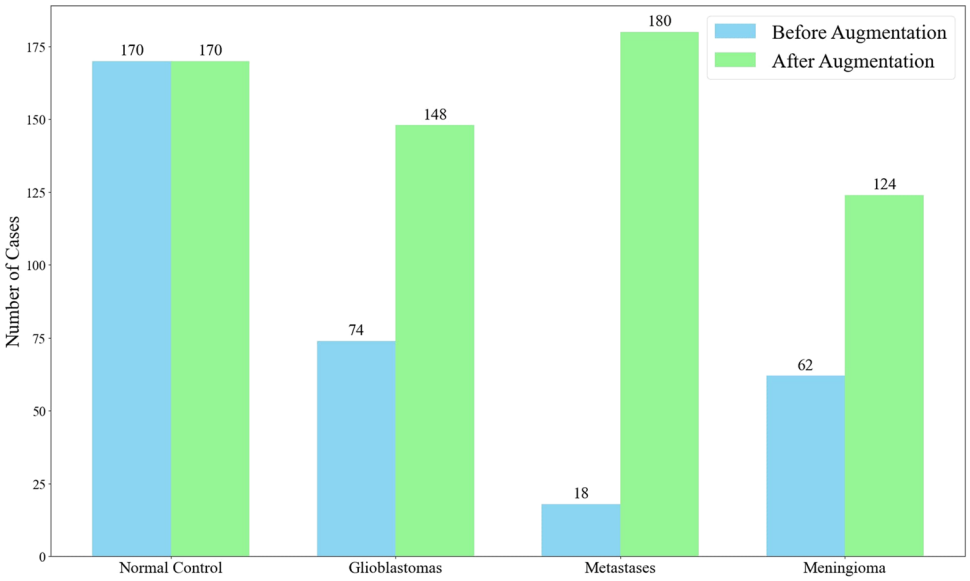


FIGURE 1 | Number of cases before and after data augmentation.

3.4 | Feature Extraction and Standardization

Radiomic features, including statistics of GLCM and Curvelet coefficients, were extracted from preprocessed images. All feature extraction was implemented using Python (3.10.0), leveraging libraries such as pydicom, skimage, NumPy, and custom Python code.

3.4.1 | Gray Level Co-Occurrence Matrix (GLCMs)

GLCM coefficients prepare a second-order approach for generating texture features of images. These coefficients show the conditional joint probabilities of all pairs with combinations of gray levels in the spatial window of interest, given two parameters: interpixel distance (δ) and orientation (θ). The probability measure is stated as:

$$P_r(x) = \{C_{ij}|(\delta, \theta)\} \tag{1}$$

Where C_{ij} (the co-occurrence probability between gray levels i and j) is determined as:

$$C_{ij} = \frac{P_{ij}}{\sum_{i,j=1}^{G} P_{ij}} \tag{2}$$

where P_{ij} indicates the number of occurrences of gray levels i and j within the given window, given a certain (δ, θ) pair; and G is the quantized number of gray levels. The sum in the denominator illustrates the total number of gray level pairs (i, j) within the window [6].

Python's skimage.feature module was used to compute the GLCMs through the greycomatrix and greycoprops functions. For each image, GLCM coefficients were extracted in the four directions (0°, 45°, 90°, and 135°) with an interpixel distance of 1. Subsequently, multiple texture features based on Haralick's method [31] were generated. The radiomics in four directions were averaged to provide a unique feature vector. Generated GLCM-based features included energy or angular second moment (ASM), contrast, correlation, cluster prominence, cluster shade, dissimilarity, entropy, homogeneity, variance, and max probability (Supporting Information S1: Table S1).

3.4.2 | Curvelet Coefficients

The Curvelet transform efficiently represents curves, lines, and edges by decomposing images into sub-images and applying the Ridgelet transform to each [14, 32, 33]. It uses a pair of windows, including W(r), a radial window, and V(t), an angular window, where r and θ are polar coordinates in the frequency domain (Equations 3 and 4). This transform extracts detailed multi-scale texture features, capturing both large-scale and fine-scale information in images.

$$\sum_{j=-\infty}^{\infty} \mathrm{W}^2(2^j r) = 1 \, for \quad r \in \left(\frac{3}{4}, \frac{3}{2}\right) \tag{3}$$

$$\sum_{j=-\infty}^{\infty} \mathrm{v}^2(\mathrm{t} - 1) = 1 \, for \quad t \in \left(-\frac{1}{2}, \frac{1}{2}\right) \tag{4}$$

A "polar wedge," denoted by U_j, is supported by the radial and angular windows W and V. In the Fourier domain, U_j is defined by Equation 5:

$$U_j(r, \theta) = 2^{\left(-\frac{3j}{4}\right)} W(2^{-j} r) V\left[\frac{2^{\left(\frac{j}{2}\right)}\theta}{2\pi}\right] \tag{5}$$

The Curvelet transform can be defined as a function of $\mathrm{x} = (\mathrm{x}_1, \mathrm{x}_2)$ at scale $2^{-\mathrm{j}}$, direction θ_l, and position $\mathrm{x}_k^{(j,l)}$ as shown by Equation 6:

$$\varphi_{j,l,k}(x) = \varphi_j\left(R_{\theta_l}\left(x - x_k^{(j,l)}\right)\right) \tag{6}$$

where R_θ represents the rotation in radians.

In this study, Curvelet coefficients were extracted using Python with the PyCurvelab library. Each MR image underwent a 4-level Curvelet decomposition, generating multiple sub-bands at each level. From these sub-bands, we derived statistics including mean, standard deviation, root mean square (RMS), kurtosis, skewness, angular second moment (ASM) or energy, sum entropy, contrast, correlation, sum of squares or variance, inverse difference moment (IDM), sum average, sum variance, entropy, difference variance, difference entropy, and information measure of correlation 1 and 2 (IMC1 and IMC2) (Supporting Information S1: Table S1). By incorporating all computed statistics across sub-bands and scales, the final Curvelet-based feature vector provided a detailed and comprehensive representation of the MR images. The extracted radiomic features adhered to the definitions provided by the Image Biomarker Standardization Initiative (IBSI) [34]. Standardization of a data set is a common requirement for many machine learning estimators. We used the StandardScaler library to apply the Z-score [35] to standardize extracted features.

3.5 | Feature Selection

To identify the most informative subset of features for machine learning (ML) classifier training, we employed the Least Absolute Shrinkage and Selection Operator (LASSO) algorithm, a linear regression method that performs both variable selection and regularization by applying an L1 penalty [19]. The regularization parameter (α) was set to 0.01 to balance model sparsity and predictive performance [36]. This process initially selected 14 features from two groups of Curvelet and GLCM features, comprised of Mean, Kurtosis, ASM, Curvelet Contrast, Sum Variance, Entropy, IMC1, and IMC2 from Curvelet-based and Contrast, Dissimilarity, Homogeneity, Energy, Variance, and Cluster Shade features from GLCM-based statistics.

To address potential multicollinearity among selected features and enhance model robustness, Spearman's rank correlation coefficient (ρ), a non-parametric test assessing monotonic relationships between variables, was computed separately for each feature group [37]. Features exhibiting high collinearity ($|\rho| > 0.70$) were excluded to ensure feature independence and reduce redundancy [38]. All analyses were implemented using Python (3.10.0).

The final retained features included Mean, Kurtosis, ASM, Contrast, Sum Variance, and IMC1 from Curvelet-based and Contrast, Dissimilarity, Energy, and Variance from GLCM-based features. This rigorous process ensured that the retained features were both nonredundant and highly informative, improving the interpretability of the results [39]. The final selected radiomic features from the GLCM and Curvelet groups were used to train classifiers both individually for each group and collectively by combining the features from both groups.

Principal component analysis (PCA) [20] was conducted as an alternative technique for reducing dimensionality, with variance thresholds set at 90%, 95%, and 99%. This method decreases the number of features while preserving the most critical information. The components chosen by each PCA model were subsequently utilized for training ML models. A visual summary of the feature extraction and selection pipeline is presented in Figure 2.

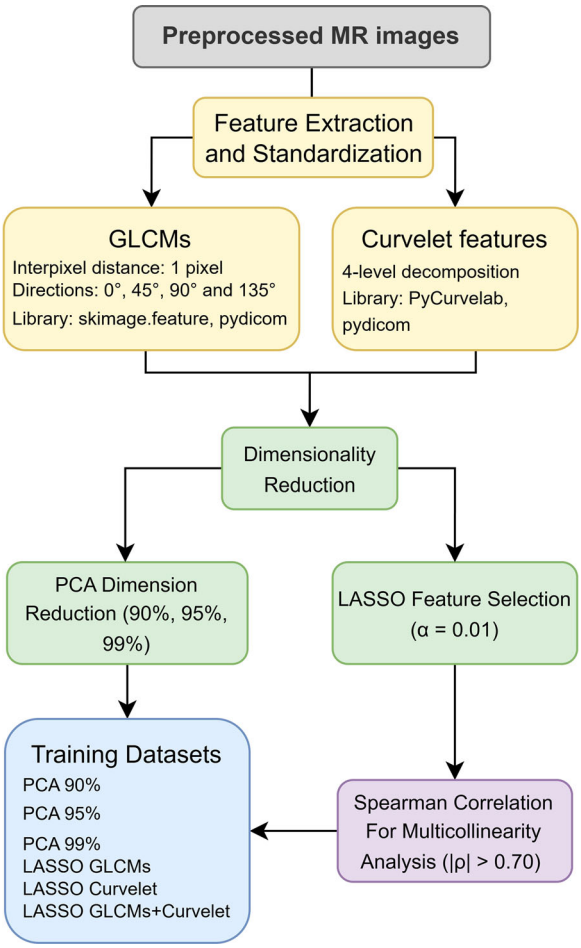


FIGURE 2 | A visual summary of the feature extraction and selection pipeline.

All steps were performed in Python (3.10.0) using open-source libraries such as NumPy, Pandas, and scikit-learn to ensure reproducibility, efficiency, and seamless integration throughout the analysis workflow.

3.6 | Machine Learning Models Training and Evaluation

The data set was split into training and testing subsets using a 70-30 ratio. The training subset was employed to train all investigated ML models, including SVM, Random Forest, CatBoost, LightGBM, AdaBoost, KNN, Logistic Regression, XGBoost, and Naive Bayes. Model performance was subsequently evaluated on the remaining 30% hold-out test subset, utilizing accuracy, precision, recall, F1-score, and area under the curve (AUC). The used hyperparameters for these classifiers are presented in Table 1. All steps for model training and evaluations were performed in Python (3.10.0).

It should be acknowledged that to ensure a well-rounded and comprehensive evaluation, we selected a diverse set of machine learning algorithms commonly applied in radiomics-

TABLE 1 | Hyperparameters of trained machine learning models.

	ML classifier	Hyperparameters
1	SVM	kernel = 'rbf'
2	Random forest	n_estimators=200
3	LightGBM	learning_rate=0.1, max_depth=6, n_estimators=200
4	Adaboost	n_estimators=200, learning_rate=0.1
5	XGBoost	learning_rate=0.1, max_depth=6, n_estimators=200
6	KNN	K = 10
7	LR	penalty = 'l2'
8	CatBoost	iterations=200, learning_rate=0.1, depth=6
9	Naive Bayes	GaussianNB

Abbreviations: CatBoost, categorical boosting; KNN, K-nearest neighbors; LR, logistic regression; RF, random forest; SVM, support vector machine; XGBoost, extreme gradient boosting.

based classification. Our choice spans several methodological categories: ensemble methods (Random Forest, CatBoost, XGBoost, LightGBM, and AdaBoost) known for their effectiveness in handling high-dimensional data; distance-based approaches like KNN; probabilistic models such as Naive Bayes; and linear classifiers including SVM and Logistic Regression [22, 40, 41]. By incorporating representative algorithms from these varied families, we aimed not only to assess overall classification performance but also to explore which modeling approach is most suitable for our multi-class brain tumor classification task.

The hyperparameters used with each machine learning classifier were chosen following conventional practices and initial experiments with minor tuning to optimize classification performance. In particular, Random Forest used 200 decision trees (n_estimators=200), and CatBoost used 200 iterations (iterations=200) with a learning rate of 0.1 and tree depth 6. LightGBM and XGBoost were set similarly with 200 estimators (n_estimators=200), learning rate 0.1, and maximum tree depth 6. AdaBoost used 200 estimators with a learning rate of 0.1. SVM used the radial basis function kernel (kernel = 'rbf'). KNN classifier used k = 10. Logistic Regression used L2 regularization (penalty = 'l2'), and Naive Bayes used the Gaussian Naive Bayes algorithm without further parameter optimization.

To address class imbalance in the data set, the distribution of instances across tumor groups and control stages was examined. An initial imbalance was observed, particularly due to the lower number of brain metastasis samples. To mitigate this, data augmentation techniques were applied to the underrepresented categories. Despite this correction, some imbalance remained; therefore, the performances of classifiers were assessed using a comprehensive set of metrics, including precision, recall (sensitivity), specificity, F1-score, and AUC-ROC, which provide a more balanced evaluation than accuracy alone [42]. Figure 3 shows the study workflow.

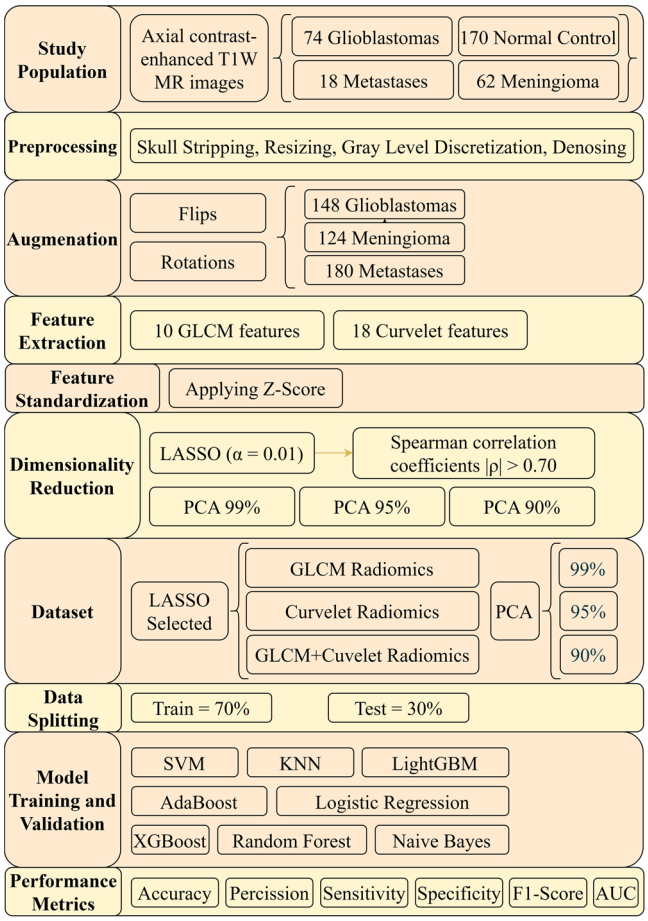


FIGURE 3 | Proposed research framework for the classification of GBM, meningioma, brain metastases, and NC cases based on radiomic features. This framework outlines the entire process, from data collection and preprocessing to radiomic feature extraction, feature standardization and selection, and machine learning model evaluation.

4 | Results

C-MRI sequences, including T1 axial with contrast for a sample case from each group of evaluated cases (NC, GBM, meningioma, and brain metastases) are depicted in Figure 4a–d, respectively. Table 2 summarizes the demographics of the study samples. Supporting Information S1: Table S1 presents all extracted GLCM and Curvelet-based features from MR images. Figure 5 shows the correlation coefficients among LASSO-selected GLCM and Curvelet-based radiomics and the final radiomics selected from each group of features. The performances of nine applied machine learning models trained on LASSO-selected features from GLCM and Curvelet-based radiomics and PCA 99% components are reported in Table 3. The performances of the trained models obtained using PCA retaining 90% and 95% of the variance and features selected by LASSO from GLCM and Curvelet features separately are detailed in Supporting Information S1: Table S2 of the supplementary file. The receiver operator curves for trained models are illustrated in Supporting Information S1: Figure S1. Figure 6 indicates the accuracy rates obtained when trained classifiers on retained components by PCA using variances of 90, 95, and 99%.

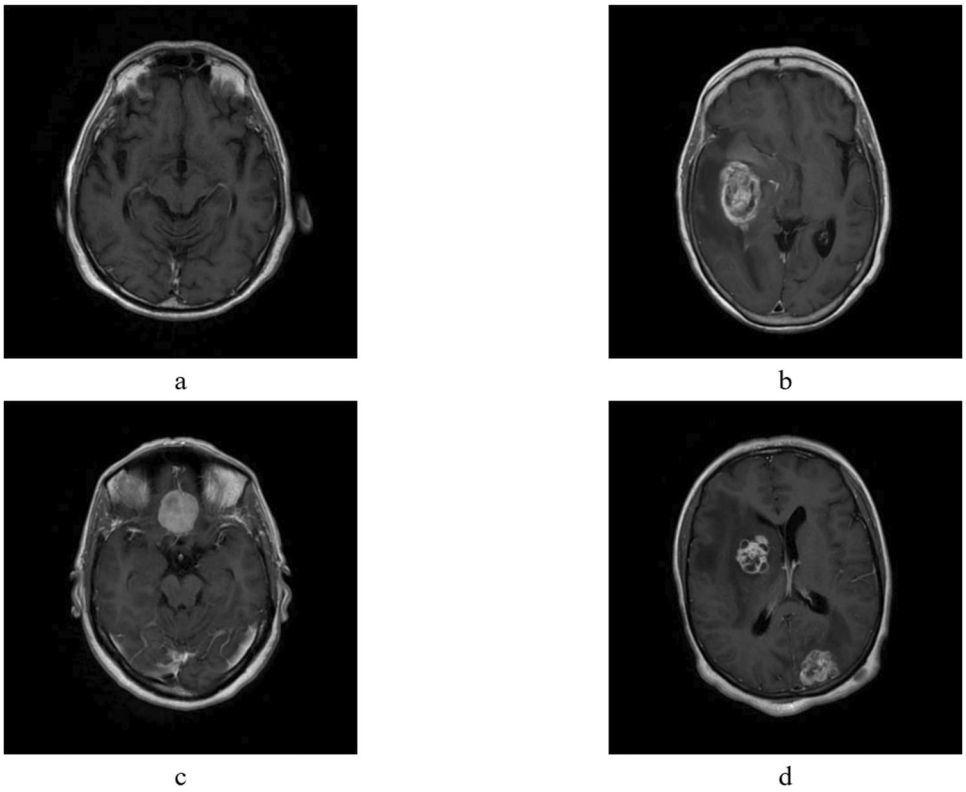


FIGURE 4 | C-MRI sequences including T1 axial with contrast for a sample case from each group of evaluated cases (a: normal control (NC), b: glioblastomas (GBM), c: meningioma, and d: brain metastases).

TABLE 2 | Demographic characteristics of 154 patients and 170 normal control cases involved in the study.

Normal controls	Age (year)	Mean	41.72 Y/O
		Range	10–86
	Gender	Male	55 (32%)
		Female	115 (68%)
Patients	Age (year)	Mean	46.81 Y/O
		Range	3–85
	Gender	Male	64 (41%)
		Female	90 (59%)

5 | Discussion

The integration of radiomics and machine learning into medical imaging represents a transformative shift in disease diagnosis and management [43]. Radiomics enables the extraction of high-dimensional data from medical images, providing comprehensive insights into tumor heterogeneity and other pathological features that may not be discernible through traditional image analysis [44]. When combined with ML classifiers, these quantitative features can enhance the accuracy and efficiency of disease classification, leading to earlier detection, personalized treatment plans, and improved patient prognoses. Such advancements are crucial for addressing the complexities of diagnosing and treating diverse brain abnormalities [45–47]. Although this study focuses on contrast-enhanced T1-weighted MR images, which were the available sequence with our system, recent studies indicate that incorporating multiple imaging modalities can further boost diagnostic accuracy in complex medical conditions by providing richer, complementary information, thereby reducing diagnostic uncertainty and improving overall performance in medical imaging tasks [48].

In this study, radiomic features, including GLCM and Curvelet-based, were extracted. The most relevant features were selected using the LASSO algorithm with an α value of 0.01. The multicollinearity of extracted features was assessed using Spearman's correlation analysis to determine the robust features for model training. Principal components were derived using PCA with variance thresholds of 90%, 95%, and 99% for dimensionality reduction. Multiple machine-learning classifiers, including SVM, Random Forest, CatBoost, LightGBM, AdaBoost, KNN, Logistic Regression, XGBoost, and Naive Bayes, were trained accordingly to achieve optimal classification of glioblastoma, meningioma, brain metastases, and normal control cases. Many studies highlight the efficacy of GLCM alone, particularly for encoding local texture information, while others demonstrate strong classification metrics using wavelet, Curvelet, or deep CNN features [1, 3, 49]. Our strategy combined GLCM and Curvelet feature sets, which are subsequently optimized through feature selection via LASSO and dimensionality reduction via PCA, an approach that has proven beneficial in reducing the dimensionality of high throughput radiomic features [50–52].

The final retained features, following multicollinearity analysis, included mean, kurtosis, energy (ASM), contrast, sum variance, and IMC1 from Curvelet-based features, along with contrast,

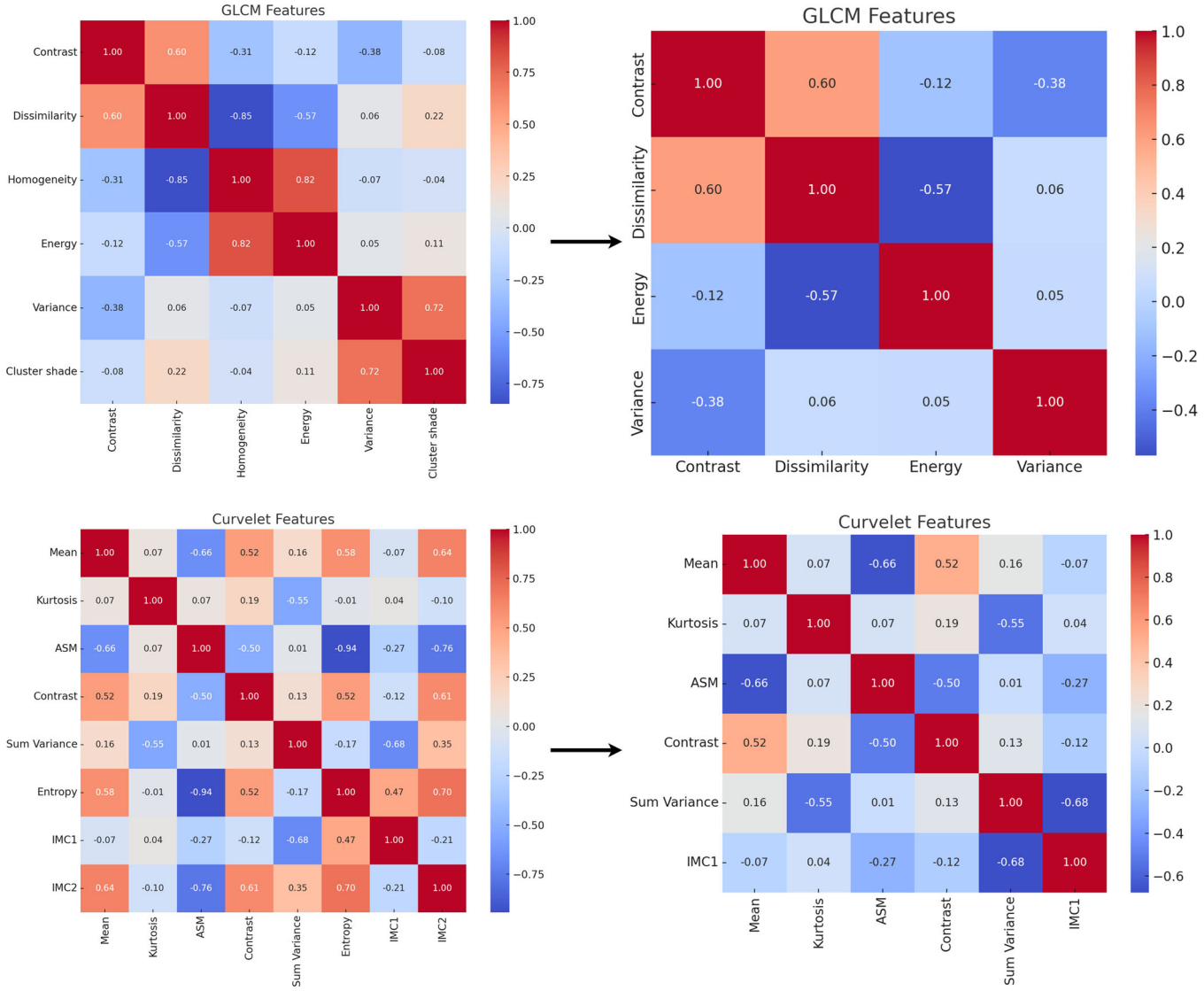


FIGURE 5 | Correlation coefficients among LASSO-selected GLCM and Curvelet-based radiomics, as well as final radiomics selected from each group of features (ASM: angular second moment, IMC1: information of measure of correlation 1).

dissimilarity, energy, and variance from GLCM-based features. This careful selection process ensured that only highly informative and nonredundant features were used, thereby improving the interpretability of the results [39]. The radiomic features extracted from MR images provided valuable insights into the textural characteristics of different brain tumor types and normal brain tissues. Contrast, which measures local gray-level variations, captures intensity differences between a reference pixel and its neighbors, providing essential texture information [53, 54]. Dissimilarity, an indicator of the distance between pixel pairs, reflects the mean difference in intensity distribution, with larger values signifying greater disparities among neighboring pixels [55]. Energy or ASM quantifies voxel intensity magnitudes and uniformity, ranging from 0 to 1, with higher values indicating more homogeneous regions [56]. Variance, or sum of squares, represents inhomogeneity and dispersion, with higher values denoting a wider spread of gray levels [57]. The information measure of correlation I (IMC1) serves as an indicator of image texture complexity [58]. Sum variance, incorporating all variances, further enhances the representation of textural differences [59]. Kurtosis, which describes the "tailedness" of pixel intensity distribution, helps characterize structural and textural changes in an image. A positive kurtosis (leptokurtic) suggests a sharper peak and heavier tails, while a negative kurtosis (platykurtic) implies a flatter distribution. A mesokurtic distribution (kurtosis = 0) closely resembles a normal distribution, making kurtosis a valuable measure for assessing textural patterns [60, 61]. Mean is an indicator of the average intensity of pixel values in an image, reflecting the overall brightness and aiding in the assessment of general intensity distribution [62]. By leveraging these selected features, our approach provides a robust framework for distinguishing among different brain tumor types, enhancing 4-way classification accuracy and reliability.

Based on the findings, Random Forest and CatBoost achieved the highest classification performances, with accuracy rates of 95.20% when trained on LASSO-selected features and 92% with PCA99% retained variance, respectively. These models exhibited excellent sensitivity (95.2%), specificity (98.4%), and AUC scores (higher than 0.99), underscoring their robustness in differentiating among tumor types. Further evaluation using

TABLE 3 | The performances of applied classifiers trained on LASSO-selected features from both GLCM and Curvelet-based radiomics and the PCA 99% method.

Classifier	Training data set	Accuracy (%)	Precision (%)	Sensitivity (%)	Specificity (%)	F1-Score (%)	AUC (%)
SVM	LASSO	84	85.55	84	94.67	83.98	97.17
	PCA 99%	85.6	87.41	85.6	95.2	85.68	98.99
Random forest	LASSO	95.2	95.27	95.2	98.41	95.2	99.52
	PCA 99%	92	92.07	92	97.34	91.95	99.14
CatBoost	LASSO	95.2	95.27	95.2	98.41	95.2	99.56
	PCA 99%	92	92.25	92	97.35	91.98	99.43
LightGBM	LASSO	92.8	92.98	92.8	97.62	92.83	99.46
	PCA 99%	90.4	90.7	90.4	96.8	90.34	99.19
AdaBoost	LASSO	72	73.04	72	90.62	71.9	87.77
	PCA 99%	64.8	65.51	64.8	88.31	65.01	85.57
KNN	LASSO	90.4	91.04	90.4	96.79	90.26	97.9
	PCA 99%	89.6	90.17	89.6	96.52	89.46	96.98
Logistic regression	LASSO	64	63.73	64	88	63.72	86.24
	PCA 99%	64.8	66.9	64.8	88.29	65.44	86.27
XGBoost	LASSO	93.6	93.89	93.6	97.88	93.62	99.51
	PCA 99%	90.4	90.6	90.4	96.81	90.35	99.18
Naive bayes	LASSO	57.6	56.94	57.6	85.82	55.13	84.25
	PCA 99%	70.4	72.56	70.4	90.12	70.25	87.13

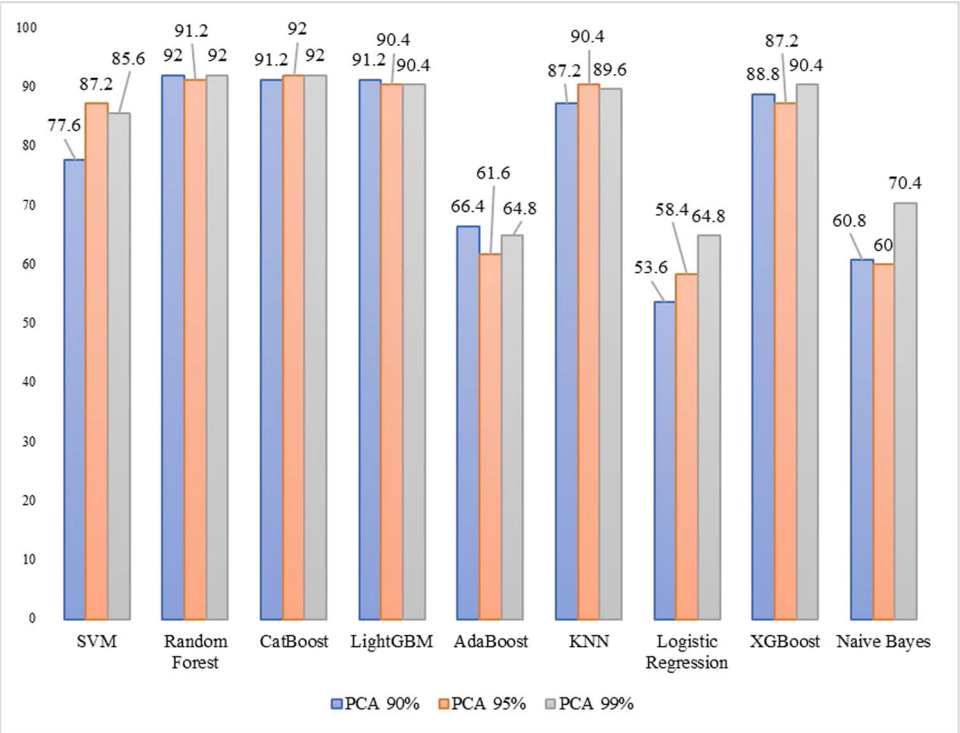


FIGURE 6 | The accuracy rates obtained when trained classifiers on retained components by PCA using variances of 90, 95, and 99%.

additional metrics, including precision and F1-score, revealed that Random Forest achieved the highest performance, excelling in both identifying true positive cases and minimizing false negatives. CatBoost followed closely, demonstrating consistently high precision and sensitivity values. SVM, KNN, and XGBoost also exhibited strong classification performances, particularly when trained on LASSO-selected features, achieving accuracies above 90%. One of the key explanations for the better performance of Random Forest and CatBoost models lies in their ensemble-based design, which combines multiple decision trees to better learn nonlinear relations. Such a design prevents overfitting by averaging out or boosting predictions of single learners, enhancing model stability [63–65]. Additionally, these algorithms are relatively insensitive to noise and outliers and can be geared for high-dimensional, high-texture datasets common in radiomics. In contrast, less complex algorithms (e.g., Logistic Regression or Naïve Bayes) tend to depend upon linearity or independence of features and therefore can't make optimum use of brain tumor images' intricate, multi-class texture patterns [63, 66]. Specifically, Naive Bayes and Logistic Regression demonstrated lower classification performance, with accuracy rates below 65%, highlighting the limitation of simpler probabilistic models when dealing with high-dimensional radiomic features.

Among other ensemble models, AdaBoost demonstrated moderate classification performance, with accuracies of 72% using LASSO-selected features and 64.8% with PCA99%. LightGBM performed notably better, achieving 92.8% accuracy with LASSO-selected features. These results further support the advantage of using LASSO in enhancing model performance and confirm that the combination of Curvelet and GLCM texture features with appropriate feature selection methods yields highly informative inputs for training machine learning models.

PCA-based dimensionality reduction, while effective, resulted in slightly lower performance compared to LASSO-selected features. It should be noted that although PCA is an unsupervised data reduction technique that preserves overall variance in data, it does not necessarily concentrate on features that are most useful for classification [20, 67]. In contrast, LASSO applies an extra penalty term to include only those features that have the greatest impact on predictive performance, essentially removing unimportant and redundant variables [68]. Such 'task-specific' concentration is probably why better results were achieved using features selected by LASSO compared to those produced by PCA in our multi-class brain tumor data set [69]. Another issue is that while using PCA, we sometimes keep components accounting for variance unrelated to the problem at hand and consequently have an inferior discriminative feature space [70].

The stability of machine learning classifiers concerning various feature extraction and dimensionality reduction techniques significantly impacts their diagnostic utility in medical imaging. Kujur et al. [71] emphasize the importance of evaluating model performance concerning data complexity, particularly in the classification of brain tumors and Alzheimer's disease using MRI. Their findings demonstrate that classifier performance is closely tied to the intrinsic complexity of the training data, suggesting that model architectures should be adapted accordingly. These insights underscore the need to assess classifiers not only through conventional performance metrics but also in the context of data complexity, thereby enhancing the clinical relevance and validity of medical image analysis systems. Furthermore, emerging approaches such as the relevance-aware capsule network (XAI-RACapsNet), which incorporates explainable AI techniques for breast cancer detection in mammography, highlight the increasing demand for transparency and interpretability in AI-driven medical tools [72]. Such methods contribute to improved clinical trust and foster broader adoption of machine learning solutions in healthcare.

The combination of GLCM and Curvelet-based features provided richer textural information, leading to relatively better classification results compared to using each feature group alone (Supporting Information S1: Table S2). Specifically, using combined radiomics of GLCM and Curvelet-based features improved the AUC for most classifiers. The most significant enhancements were observed in AdaBoost and Logistic Regression performances, where the AUC increased by approximately 12% compared to using GLCM features alone, and by around 5% and 10% compared to using Curvelet-based features, respectively. The SVM classifier also benefited notably, with an 8.33% increase in AUC over GLCM-based training and around a 1% improvement over Curvelet-based training. Random Forest, CatBoost, and XGBoost exhibited moderate gains, with AUC improvements of approximately 2%, 2%, and 1.5%, respectively, when trained on combined features compared to GLCMs alone. LightGBM and KNN demonstrated smaller but consistent improvements, with AUC increases of around 1% and more than 2.5%, respectively, over GLCM-based training. The Naive Bayes model also showed an increase in AUC of approximately 7.5% compared to GLCM-based training and less than 2% compared to Curvelet-based training. These findings highlight the synergistic effect of integrating GLCM and Curvelet features, which capture complementary aspects of textural information, thereby enhancing the discriminatory power of classifiers in brain tumor classification.

Findings (Figure 4) demonstrate that Random Forest and CatBoost models exhibited consistently high accuracy across all PCA levels (90, 95, and 99%), with values ranging between 91.2% and 92.0%, indicating their robustness to feature dimensionality variations. LightGBM also maintained relatively stable performance, albeit slightly lower, with a maximum accuracy of 91.2% at PCA 90% and a slight decrease to 90.4% at PCA 95% and 99%. XGBoost demonstrated strong performance, achieving its highest accuracy (90.4%) at PCA 99%, which is comparable to LightGBM. SVM showed a significant improvement when increasing PCA from 90% (77.6%) to 95% (87.2%) but experienced a slight decline at PCA 99% (85.6%), suggesting a potential trade-off between dimensionality reduction and feature retention. KNN displayed a similar trend, with its peak accuracy at PCA 95% (90.4%) before decreasing slightly at PCA 99% (89.6%). In contrast, AdaBoost and Logistic Regression showed weaker performance, with AdaBoost peaking at 66.4% (PCA 90%) but declining to 61.6% at PCA 95% before slightly recovering at PCA 99% (64.8%). Logistic Regression improved progressively from PCA 90% (53.6%) to PCA 99% (64.8%), indicating a positive effect of increased feature retention. Naïve Bayes showed noticeable improvement at PCA 99% (70.4%), outperforming its results at PCA 90% (60.8%) and PCA 95%

(60%), suggesting that retaining more principal components enhances its classification capability [70, 73]. Using PCA with a 90% variance threshold reduced the data set to the most informative principal components, effectively balancing dimensionality reduction and classification performance. While models such as Random Forest, CatBoost, and LightGBM maintained high accuracy levels, others, such as SVM and Naive Bayes, exhibited significant variations in performance, indicating their sensitivity to the retained feature set. As the variance threshold increased to 95%, additional components were preserved, leading to improvements in several models, particularly KNN and XGBoost, which benefited from enhanced feature representation [70, 74]. At the 99% variance threshold, the retained components allowed a more detailed analysis of complex patterns, resulting in stable or improved performance for most classifiers. However, the observed variations across PCA levels highlight a trade-off between dimensionality reduction and model complexity, where excessive feature retention may introduce redundant information, leading to overfitting in certain classifiers. These findings emphasize optimizing PCA thresholds to balance computational efficiency and classification accuracy [75–77].

5.1 | Ablation Study

To elucidate the contribution of each pipeline component, we carried out a systematic ablation study (Table 4) in which GLCM or Curvelet textures and the LASSO selector were individually removed or replaced (refer to Supporting Information S1: Table S2 to see all whole model performances trained on different datasets). Omitting Curvelet descriptors and relying solely on GLCM lowered Random Forest accuracy from 95.2% to 88.8%, while using only Curvelet features produced a smaller decline to 94.4%, confirming that the two texture families provide complementary information. Replacing the task-specific LASSO selector with unsupervised PCA (retaining 90%–99% variance) cut accuracy by 3% to 4% points despite comparable AUCs, underscoring the value of sparsity-driven feature curation for high-throughput radiomics. Even so, Random Forest maintained $\geq 91\%$ accuracy across all PCA settings, demonstrating resilience to moderate dimensionality shifts. Collectively, these findings show that fusing GLCM and Curvelet textures with the LASSO feature selection algorithm and addressing multicollinearity of selected features are essential for maximizing diagnostic performance in four-class brain-tumor classification.

5.2 | Comparison With the Literature

Table 5 summarizes the findings of the most relevant and recent studies on brain tumor classification, although most of the published literature focuses on binary classifications. These studies report widely varying sample sizes and imaging sequences, with some studies employing large datasets to train deep learning architectures, while others rely on smaller, specialized collections. Such variations impact the generalizability of the proposed methods, as larger data sets offer greater statistical power but also demand higher computational resources [23, 25, 78]. Furthermore, interpretability and computational feasibility play critical roles in clinical adoption, which is why several prior studies have emphasized the utility of traditional algorithms like SVM, KNN, or LR on handcrafted features [3, 49, 79]. Specifically, Malakouti et al. [23] reported an accuracy of 95.7% using a LightGBM model trained on GLCM features for the binary classification of NC vs. brain tumors. Rehman Khan et al. [24] achieved a similar accuracy of 95.1% with an ensemble model combining RF, SVM, and XGBoost, trained on features extracted from DenseNet169. Celik et al. [25] reported an accuracy of 97.2% using a KNN model trained on deep features extracted with a customized CNN algorithm. Additionally, Cepeda et al. [22] attained a classification accuracy of 94% with a Naïve Bayes algorithm for predicting meningioma consistency in preoperative surgical planning. Priya et al. [27] achieved an accuracy of 89.7% using an AdaBoost model trained on LASSO-selected shape and GLCM radiomics for the classification of glioblastoma vs. metastasis. Our results for the 4-way classification task are comparable to these studies.

While deep CNNs can extract highly discriminative features, they are often considered "black box" models and their training may be hampered by limited data availability, especially in specialized tumor subtypes [24–26]. By contrast, our approach incorporates LASSO-based feature selection, ensuring only the most relevant variables are retained, and employs PCA to systematically reduce redundancy while maintaining interpretability [27, 50, 80]. This framework not only yields high diagnostic accuracy but also offers a clearer rationale for how the features inform classification, a factor that can be essential when translating advanced algorithms into routine medical practice [14, 20, 50]. Notably, methods such as AdaBoost and logistic regression achieved lower accuracies in our experimentation, indicating that not all ensemble or linear classifiers benefit equally from the hybrid feature set or that their hyperparameters may require further optimization [26, 81]. In

TABLE 4 | Random forest ablation showing how texture sources and feature-selection choices affect classification accuracy.

Variant (Random forest)	Texture source	Dimension reduction step	Accuracy (%)	Δ vs. full (%)	AUC (%)
Full pipeline	GLCM + Curvelet	LASSO (+ ρ-filter)	95.2	—	99.52
Remove Curvelet	GLCM only	LASSO (+ ρ-filter)	88.8	−6.4	97.52
Remove GLCM	Curvelet only	LASSO (+ ρ-filter)	94.4	−0.8	99.17
Replace LASSO → PCA (99%)	GLCM + Curvelet	PCA 99%	92	−3.2	99.14
Replace LASSO → PCA (95%)	GLCM + Curvelet	PCA 95%	91.2	−4.0	99.33
Replace LASSO → PCA (90%)	GLCM + Curvelet	PCA 90%	92	−3.2	99.02

TABLE 5 | Summary information of published literature on brain tumor classification.

Study	Sample size	Sequence	Features	Feature selection	Classes	Classifier	Accuracy (%)	Precision (%)	Sensitivity (%)	Specificity (%)
Malakouti et al.	3762	—	GLCM	—	NC and Tumor	LightGBM	95.7	96.2	95.5	—
						AdaBoost	—	95.0	99.6	95.5
						Gradient Boosting	—	91.3	99.1	95.5
						RF	—	93.3	99.1	95.5
						QDA	—	97.0	100.0	68.2
						LDA	—	80.0	99.1	80.0
						LR	—	98.3	98.7	81.8
			—			Transfer Learning	99.3	99+	99+	—
Rehman Khan et al.	3064	T1	DenseNet169	—	NC and Tumor	RF	91.0	91.0	91.0	90.0
						SVM	92.0	92.0	92.0	90.0
						XGBoost	94.0	94.0	94.0	93.0
			Combination of RF, SVM, and XGBoost			Hybrid-NET	95.1	95.0	95.0	94.0
Celik et al.	7023	T1	DenseNet201	—	NC and Tumor	SVM	97.0	97.0	97.0	—
						KNN	96.7	97.0	97.0	—
			EfficientNetB0			SVM	97.9	98.0	98.0	—
						KNN	97.0	97.0	97.0	—
			Proposed CNN Model			SVM	93.9	94.0	94.0	—
						KNN	97.2	97.0	97.0	—
Cinarer et al.	1980	—	REMBRANDT data set	—	NC and Tumor	SVM	90.0	90.0	90.0	—
						KNN	87.0	86.7	86.0	—
						LDA	83.0	83.3	83.0	—
						RF	83.0	83.3	83.0	—
Kibriya et al.	15320	T1	ResNet-18	—	NC and Tumor	ResNet-18	97.8	98.3	98.0	—
			GoogLeNet			GoogLeNet	97.4	97.3	97.3	—
			—			ResNet-18 + SVM	98.0	98.3	98.0	—
						GoogLeNet + SVM	97.6	97.3	97.3	—
Gupta et al.	50	—	Curvelet Transform	—	Tumor and Nontumor	SVM	85.0	76.9	100.0	—
Gurusamy et al.	150	—	Wavelet Transform	—	NC and Tumor	SVM	98.0		—	—
Cepeda et al.	18	T1, T2 and ADC	Radiomic features	—	Meningioma type	Naive Bayes	94.0	95.0	—	—
						LR	88.8	91.4	—	—
						KNN	88.8	90.6	—	—
						Neural Network	88.9	90.6	—	—

(Continues)

TABLE 5 | (Continued)

Study	Sample size	Sequence	Features	Feature selection	Classes	Classifier	Accuracy (%)	Precision (%)	Sensitivity (%)	Specificity (%)
Chen et al.	233	T2-FLAIR	HOG	IG, ReliefF, and FCBF	NC and Tumor	SVM	86.3 (Detection), 76.3 (Grading)	—	89.4 (Detection), 83.7 (Grading)	80.5 (Detection), 68.7 (Grading)
Chen et al.	134	T1C	GLCM, NGLDM, GLZLM and GLRLM	Distance Correlation	NC, Glioma, Meningioma and Pituitary	Distance Correlation + LDA	78.0	—	69.0	86.0
						Distance Correlation + LR	79.0	—	71.0	85.0
Bahadure et al.	135	T1, T2 and FLAIR	GLCM and BWT	—	NC and Tumor	SVM	96.5	—	97.7	94.2
Artzi et al.	439	T1C	Clinical information, tumor location, first- and second-order statistical, morphological, wavelet features, and bag-of-features	—	Glioblastoma, Metastasis	SVM	75.0 to 90.0	11.0 to 100.0	76.0 to 99.0	85.0
					Four (Subtype of metastasis)		44.0 to 72.0	—	35.0 to 51.0	76.0 to 89.0
Usman et al.	30	T1, T2, T1C and FLAIR	Intensity, Intensity Differences, Neighborhood, Wavelet Texture	—	Background, necrosis, edema, enhancing and non-enhancing tumor	RF	—	—	95.0	89.0
Rinesh et al.	250	—	Hyperspectral images	Firefly	Tumor and Nontumor	k-means Clustering + MFNN	96.5	—	96.3	98.2
Priya et al.	120	T1, T2, T1C, FLAIR and ADC	Radiomic features	LASSO	Glioblastoma and Metastasis	AdaBoost	89.2	—	88.7	89.7
		FLAIR					89.7	—	91.7	87.7
		Combined T1C and FLAIR					88.8	—	90.0	87.7
Kang et al.	253	—	DenseNet-169	—	Normal, Glioma, Meningioma and Pituitary	SVM	98.0	—	—	—
	3000	—	DenseNet-121, ResNeXt-101				98.3	—	—	—
	3064	T1	DenseNet-169, ShuffleNet V2				92.0	—	—	—
Our study	622	T1	GLCM+Curvelet	LASSO	Normal, Glioma, Meningioma, and Brainmet	CatBoost	95.2	95.27	95.2	98.41

Abbreviations: Adaboost, adaptive boosting; ADC, apparent diffusion coefficient; BWT, Berkeley wavelet transformation; CNN, convolutional neural network; FCBF, fast correlation-based filter; GLCM, gray-level co-occurrence matrix; GLZLM, grey-Level zone length matrix; HOG, histogram of oriented gradients; IG, information gain; KNN, k-Nearest Neighbors; LDA, linear discriminant analysis; LightGBM, light gradient-boosting machine, LR, logistic regression; NC, normal control; NGLDM, neighboring gray-level dependence matrix; QDA, quadratic discriminant analysis; RF, random forest, SVM, support vector machine; XGBoost, extreme gradient boosting.

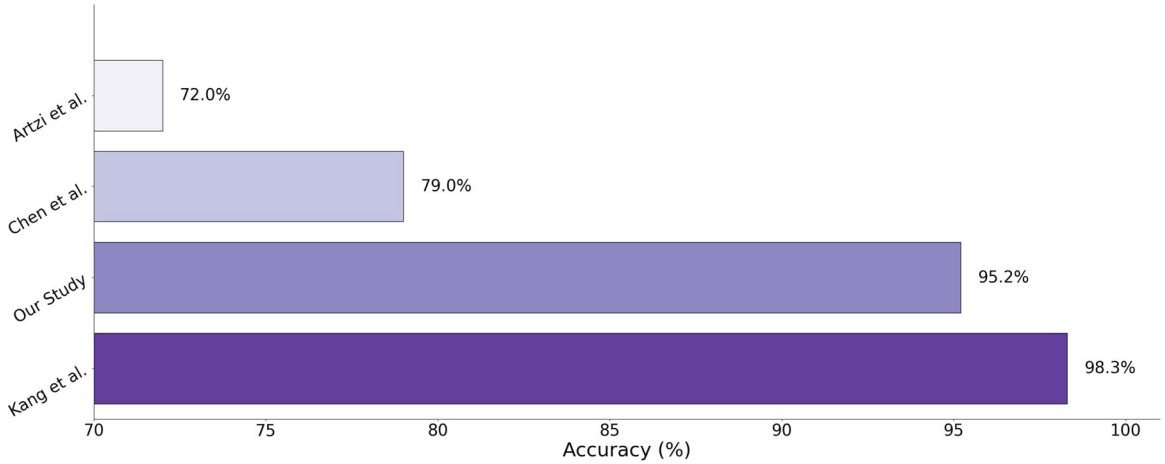


FIGURE 7 | Comparative analysis of classification accuracy between the proposed method and published machine learning approaches for multi-class brain abnormality detection.

contrast, the significantly elevated performance of CatBoost, Random Forest, and XGBoost with combined GLCM and Curvelet-based radiomics and PCA scenarios highlights the value of integrating diverse textural descriptors and dimensionality reduction [14, 56, 79]. Overall, our results align with and, in some cases, exceed the performance metrics reported in earlier works [1, 23–25, 79], reaffirming the merit of our multimodal, carefully optimized approach for brain tumor classification. The reviewed literature in Table 5 underscores the potential of DenseNet architectures and other deep learning models in tumor classification tasks [25, 26, 80]. Our findings, however, show that classical machine learning methods can closely match or surpass these scores when trained on robust, complementary feature representations, as evidenced by our Random Forest, CatBoost, and XGBoost models consistently achieving accuracy, precision, and specificity above 90% (Supporting Information S1: Table S2). Moreover, a comparative analysis of classification accuracy between the proposed methods, CatBoost and Random Forest using hybrid GLCM and Curvelet features, and previously published machine learning approaches for multi-class brain abnormality classification is presented in Figure 7. Among the reviewed studies, only the work by Kang et al. [80], which employed deep feature extraction combined with SVM classifiers, achieved a higher accuracy than our proposed approach.

5.3 | Study Limitations and Future Directions

Our study presents a robust radiomics-based machine learning approach for brain tumor classification, integrating GLCM and Curvelet-based feature extraction with advanced feature selection methods. This combination improved classification accuracy by capturing complementary texture information, demonstrating the potential of radiomics in automated brain tumor diagnosis. Additionally, the use of multiple classifiers enabled a comprehensive evaluation of different modeling strategies, reinforcing the reliability of the findings. Despite these promising results, certain limitations should be acknowledged. First, although the data set is diverse, its size remains limited compared to large-scale deep learning studies. A restricted data set size reduces statistical power and may cause models to overfit to features specific to our patient cohort, thereby limiting the generalizability of the results. This limitation could be partially addressed through data augmentation techniques; however, incorporating multi-institutional datasets or utilizing public radiomics repositories would introduce greater heterogeneity and enhance the robustness and external validity of future studies [82, 83]. Second, variations in MRI acquisition protocols may introduce inconsistencies in radiomic feature extraction, potentially affecting reproducibility. This challenge could be mitigated through standardized imaging protocols, the use of vendor-neutral imaging phantoms, or the application of domain adaptation algorithms to harmonize feature distributions across different scanners and institutions [84].

Lastly, this study primarily focused on classical machine learning models and did not explore deep learning-based feature extraction approaches, which may capture more abstract and discriminative patterns. Future work should investigate the integration of deep learning methods, such as CNNs, with handcrafted radiomic features to further improve classification performance. Additionally, incorporating clinical and genomic data could enable the development of multimodal predictive models for brain tumor prognosis and treatment response. Another important direction involves the deployment of these models into real-time clinical workflows to evaluate their practical utility in assisting radiologists. By addressing these challenges, future research can contribute meaningfully to the advancement of AI-driven diagnostic tools in neuro-oncology.

6 | Conclusion

This study developed and validated a radiomics-based machine learning framework for classifying brain tumors using texture features extracted from MRI scans. A hybrid feature extraction approach was employed, combining GLCM and Curvelet features, followed by dimensionality reduction techniques (LASSO and PCA). To enhance robustness, multicollinearity among LASSO-selected features was assessed, and redundant features were removed. Among the evaluated machine learning models, Random Forest and CatBoost outperformed others, benefiting from their ensemble-based architectures and strong capability to handle high-dimensional texture data. Ablation analysis

further demonstrated that combining radiomic features significantly improved classification accuracy. Overall, the findings meet the study's primary objectives and underscore the potential of radiomics combined with machine learning techniques to enhance diagnostic accuracy and reliability in brain tumor characterization.

Author Contributions

Mostafa Jafari: writing – review and editing, formal analysis, writing – original draft, methodology, software, data curation. **Masoumeh Rahmani:** conceptualization, writing – review and editing, methodology, supervision, software, validation. **Sanaz Alibabaie:** data curation, writing – original draft, writing – review and editing, software, formal analysis. **Jafar Fatahiasl:** conceptualization, writing – review and editing, methodology. **Sasan Razmjoo:** conceptualization, writing – review and editing, methodology. **Mohammad Javad Tahmasebi Birgani:** conceptualization, writing – review and editing, methodology. **Marziyeh Tahmasbi:** conceptualization, writing – review and editing, writing – original draft, resources, methodology, formal analysis, supervision, validation.

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Ethics Statement

The retrospective study protocol was approved by the ethics committee of Ahvaz Jundishapur University of Medical Sciences (Ethics code: IR.AJUMS.REC.1399.839). All study procedures adhered to the Declaration of Helsinki. Patient data confidentiality was strictly maintained, and all images were anonymized before use.

Conflicts of Interest

The authors declare no conflicts of interest. All authors have read and verified the final version of the manuscript. The financial support received from the Ahvaz Jundishapur University of Medical Sciences covered the initial costs of doing the research. We declare that the Ahvaz Jundishapur University of Medical Sciences (Grant No: CRC 9927), as the financial supporter of the study, was not involved in the study design, collection, analysis, and interpretation of data, writing of the report, or the decision to submit the report for publication.

Data Availability Statement

The authors confirm that the data supporting the findings of this study are available within the article and its supplementary materials. Any necessary information, such as anonymized MR images, can be provided upon request by other researchers to the corresponding author.

Transparency Statement

The lead author Marziyeh Tahmasbi, affirms that this manuscript is an honest, accurate, and transparent account of the study being reported; that no important aspects of the study have been omitted; and that any discrepancies from the study as planned (and, if relevant, registered) have been explained.

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Supporting Information

Additional supporting information can be found online in the Supporting Information section.
Supporting Figure S1: Receiver operator curves for trained models. **Supporting Table S1:** GLCM and Curvelet-based radiomics extracted from images of included glioblastomas, meningioma, brain metastases, and normal control cases. **Supporting Table S2:** The performances of trained classifiers based on LASSO-selected features and the PCA method.